DIGITAL SAT & EST MATH COURSES

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EST1 May 2026

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Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	75 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

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Practice Questions

50 QUESTIONS

1 **Question 1.** If $2x + 3y = 9a$ and $-3x + 11 = 20$, then $3a - y =$

A. -2
B. 2
C. 3
D. 6

2 Ghassan took three Chemistry tests in Term 1 and achieved an average score of 88 out of 100. In Term 2, he took two Chemistry tests, scoring 90 and 88. In Term 3, he will take one final Chemistry test. If all tests carry equal weight, what is the minimum score Ghassan must obtain so that his overall average is greater than or equal to 90?

(A) 95
(B) 96
(C) 98
(D) 99

3 **Question 3.** Which of the following is the greatest integer that satisfies the solution set of $|x + 5| > 5x$?

A. 2
B. 1
C. -1
D. -6

4

$$y = a(x - 1)(x + 4)$$

Question 4. The vertex of the graph of the equation above is $(m, 12.5)$. The value of m is:

A. -2.0
B. -1.5
C. -0.5
D. 0.5

5 **Question 5.** Given two functions f and g defined by $f(x) = 3x + 1$ and $g(x) = \frac{2x^2 - 3}{6x + 2}$, where $x \geq 0$, what is the solution of $(f \times g)(x) = \frac{1}{2}$?

A. $x = \sqrt{2}$
B. $x = \sqrt{3}$
C. $x = 2$
D. $x = \sqrt{5}$

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6

Question 6. A box contains 11 marbles: 3 blue, 4 red, and the rest are yellow. Two marbles are drawn at random without replacement. Given that at least one of the drawn marbles is yellow, what is the probability that both drawn marbles are yellow?

A. $\frac{3}{55}$

B. $\frac{6}{55}$

C. $\frac{3}{17}$

D. $\frac{2}{7}$

7

Question 7. Given a function f defined by $f(x) = \frac{2x+5}{3x-6}$, which of the following statements is/are true about the graph of f?

I. It has a vertical asymptote at $x = 2$.

II. It has an oblique asymptote at $y = \frac{2}{3}x$.

III. It does not have a horizontal asymptote.

A. **I** only
B. **III** only
C. **I** and **III** only
D. **I**, **II**, and **III**

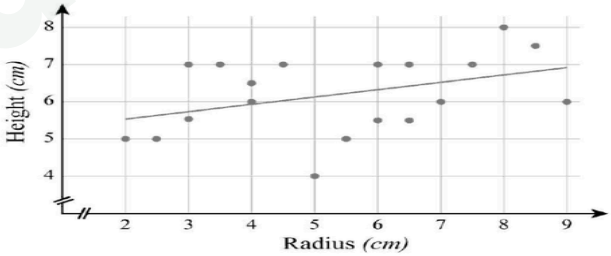
8

Question 8. If $f(x) = 3x^2 - 4x + 1$, $g(x) = 3x + 4$, and $h(x) = x^2 + \frac{2}{3}x$ what is the value of $(f + g)(h(3))$?

A. 320
B. 333
C. 340
D. 357

9

Questions 9 and 10 refer to the information below:



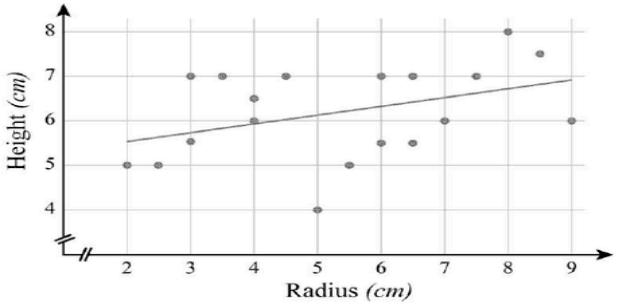
A group of students sketched several cylindrical shapes with varying radii and heights, as shown in the scatterplot above. A line of best fit is displayed.

Question 9. Which of the following equations represents the line of best fit shown above?

A. $y = 0.1915x + 5.1561$
B. $y = 0.20x - 5.10$
C. $y = 0.20x + 2.10$
D. $y = 5.1561x + 0.1915$

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10



Question 10. Another student decided to sketch a cylinder with a radius of 11 cm, following the line of best fit shown. What would be the corresponding height of the cylinder?

A. 56.9 cm
B. 7.26 cm
C. 4.30 cm
D. 2.90 cm

11

Question 11. 378 is 18% of:

A. 1,800
B. 1,950
C. 2,100
D. 2,250

12

$$\begin{cases} 3x - y = 15 \\ 6x + 2y = 18 \end{cases}$$

Question 12. The solution of the system of equations above can be represented by the pair (a, b), where a and b are two real numbers. What is the value of b?

A. -4
B. -3
C. 3
D. 4

13

Which of the following statements is true?

A. Selling the same number of keyboards in both years impacted the overall sales in 2023.
B. The bar graph shows a general decrease in sales from 2022 to 2023.
C. The bar graph shows a general increase in sales from 2022 to 2023.
D. The bar graph shows no general change in sales between both years.

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14

Questions 14 - 16 refer to the information below:

Score n	$0 \le n < 10$	$10 \le n < 14$	$14 \le n < 17$	$17 \le n \le 20$
Number of Students	7	13	11	5

A group of students at a learning center took a literature exam, and the scores were given out of 20, with a passing score of 10. The scores are summarized in the table above.

Question 14. Which of the following statements is true about the data collected?

A. The majority scored between 17 and 20.
B. The majority passed the literature exam.
C. Less than 15% of the students failed the literature exam.
D. As the scores are increasing, the number of students is decreasing.

15

Questions 14 - 16 refer to the information below:

Score n	$0 \le n < 10$	$10 \le n < 14$	$14 \le n < 17$	$17 \le n \le 20$
Number of Students	7	13	11	5

Question 15. The teacher bought chocolate bars for each student who scored 14 or more on this exam. If each chocolate bar costs \$3.11, how much did the teacher pay?

A. \$15.55
B. \$34.21
C. \$40.87
D. \$49.76

16

Questions 14 - 16 refer to the information below:

Score n	$0 \le n < 10$	$10 \le n < 14$	$14 \le n < 17$	$17 \le n \le 20$
Number of Students	7	13	11	5

Question 16. The mean of the data is found by multiplying the midpoint of each class by its frequency, adding these products, then dividing the total by the sum of the frequencies. What is the mean of the scores recorded?

A. 11.78
B. 12.61
C. 12.81
D. 13.05

17

x and y are two even natural numbers. Which of the following can be an odd integer less than x?

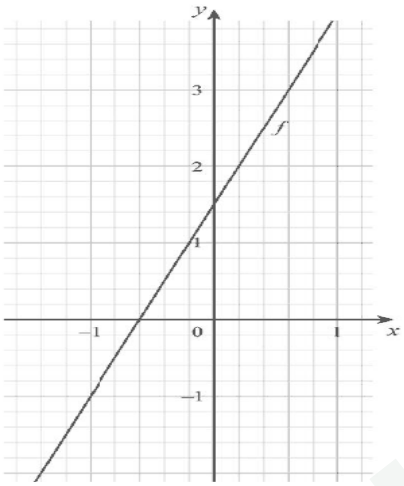
A $\frac{x+y}{2}$
B $2x - y$
C $x + \frac{y}{2}$
D $x + y$

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18 If $3r + \frac{x+r}{2} = 5(3y - r)$, what is r in terms of x and y?

A) $\frac{30y-x}{17}$
B) $\frac{x-30y}{17}$
C) $\frac{x}{17} + 30y$
D) $\frac{x+30y}{17}$

19 Consider the graph of the linear function f shown above. If the graph of $g(x) = -2x + \frac{1}{2}$ is represented on the same coordinate plane, the point of intersection of both graphs is:



A) $(-2, \frac{17}{18})$
B) $(-2, -\frac{9}{2})$
C) $(-\frac{2}{7}, -\frac{9}{2})$
D) $(-\frac{2}{9}, \frac{17}{18})$

20 If $i = \sqrt{-1}$, which of the following is/are equivalent to $A = 2i(i - 3)(3i - 1)$?

I. $A = 2i(-10i)$
II. $A = 2i(-6 - 10i)$
III. $A = 20$

A) I only
B) I and II
C) I and III
D) None of the statements is equivalent to A.

21 Which of the following is not correct regarding the graph of f with equation $f(x) = \sqrt{3x - 9}$?

A) It has a vertical asymptote at $x = 3$.
B) It passes through $(3, 0)$.
C) Its domain is all real numbers greater than or equal to 3.
D) Its range is $[0, +\infty)$.

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22 What is the local minimum of the graph of $g(x) = 2|2x - 1|$?

(A) $(1, 0)$
(B) $(0.5, 0)$
(C) $(0, 0)$
(D) $(-0.5, 0)$

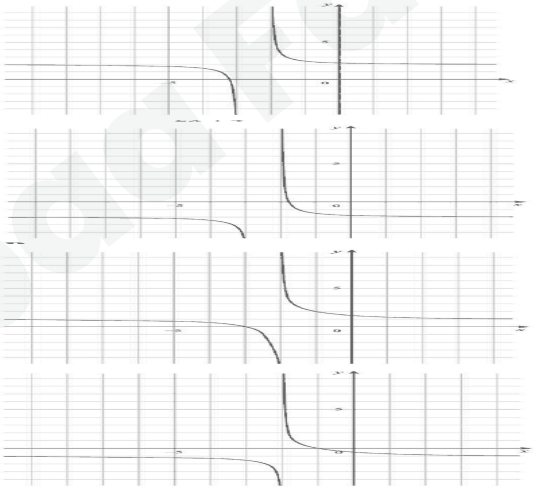
23 A square cardboard has a perimeter of 200 cm. Squares with sides of length x cm are cut out from each corner, and the remaining sides are folded up to form an open-top box of volume $5,808$ cm^3. What can be the surface area of the resulting box?

(A) $1,936$ cm^2
(B) $2,066$ cm^2
(C) $2,464$ cm^2
(D) $2,808$ cm^2

24 **Question 24.** If $\frac{3}{x+5} + \frac{7}{2x-1} = \frac{ax+b}{(x+5)(2x-1)}$ for every $x > \frac{1}{2}$, where a and b are two constants, what is the value of $a + b$?

A. 19
B. 32
C. 39
D. 45

25 Which of the following represents the graph of $f(x) = \frac{2}{2x+4} + 1$?



(A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D

26 **Question 26.** What is the largest possible integer in the solution set of $3(2x + 1) < 4(x + 1)$?

A. 3
B. 1
C. 0
D. -1

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27 The sum of two adjacent angles is $66°$. If one angle measures $(x + 3)°$ and the other is twice the first, the value of x is:

(A) 6
(B) 9
(C) 14
(D) 19

28 Given the equation $2x^2 + ax + 1 = 0$, where a is a real number, the sum of the solutions of the equation, x_1 and x_2, is -4. If $x_1 < x_2$, which of the following is x_1?

(A) $x_1 = \frac{-4-\sqrt{14}}{2}$
(B) $x_1 = \frac{-2-\sqrt{14}}{2}$
(C) $x_1 = \frac{-4-\sqrt{14}}{4}$
(D) $x_1 = \frac{-4+\sqrt{14}}{4}$

29 If y varies inversely as x^2 and $y = 12$ when $x = 5$, what is the value of y when $x = 4$?

(A) 18.25
(B) 18.75
(C) 19.0
(D) 19.75

30 Consider a circle C with radius equal to 4 cm. $\overset{\frown}{MN}$ is an arc of the circle with a length of 2π cm. What is the area of this arc sector?

(A) $\frac{\pi}{2}$ cm^2
(B) π cm^2
(C) 3π cm^2
(D) 4π cm^2

31 **Questions 31 and 32 refer to the information below:**

Tutor	$35 per hour
Saleswoman	$300 per month + $4.5 per each item sold

Brenda works two jobs: as a saleswoman at Store X from 9: 00 AM to 4: 00 PM and as an English tutor in the afternoon. Her wages are shown in the table above.

Question 31. Last month, Brenda tutored three students for 5 hours each and sold 30 items at Store X. What is the total wage Brenda is expected to receive?

A. $540
B. $610
C. $660
D. $960

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32

Questions 31 and 32 refer to the information below:

Tutor	\$35 per hour
Saleswoman	\$300 per month + \$4.5 per each item sold

Question 32. If Brenda tutors one student for 2 hours per day during the month of July and wants to earn a total of \$2,740 by the end of the month, how many items must she sell at Store X?

A. 30
B. 45
C. 60
D. 70

33

Question 33. Given the points $S(9,-3)$ and $T(12,a)$, where a is a real number, and knowing that $N(10.5,-4)$ is the midpoint of $\overline{ST}$, then the length of $\overline{ST}$ is equal to:

A. $\sqrt{10}$ units
B. $\sqrt{13}$ units
C. $\sqrt{17}$ units
D. $2\sqrt{5}$ units

34

Given the points $A(-3,4)$, $M(3,10)$, and $H(2,b)$, where b is a real number, if H lies on the line AM, then the value of b equals:

(A) 9
(B) 6
(C) 5
(D) 2

35

$$3,5,5,7,11,13,19,19,19,20,22,25,30$$

Question 35. For the set of data above, m is the mean, n is the mode, and a is the range. What is the value of $26mn-13a$?

A. 7,173
B. 7,524
C. 7,605
D. 7,875

36

Questions 36 and 37 refer to the information below:

A school has 2,400 students. A random sample of 100 students is chosen. In this sample:

- 4 students are in their final year.
- 20 students are aged between 10 and 12 years.
- The remaining students are in kindergarten.

Question 36. If the sample represents the actual repartition of students, how many of the registered students in the school are presumably in their final year?

A. 90
B. 96
C. 98
D. 110

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38 Bashar travelled to Dubai for 5 days. He had $\$3,400$ at the beginning of the trip and returned home with $\$150$ left. On average, how much money did he spend per day?

(A) $550
(B) $650
(C) $700
(D) $750

39 If $45\ \text{km/h} = a\ \text{m/min}$, what is the value of a?

(A) 75
(B) 600
(C) 750
(D) 920

40 Which of the following statements is/are true regarding the graph of the inequality $y + 3x \leq -2$?

I. It represents the region above the line of equation $y = -3x - 2$.
II. It passes through $(1, -5)$.
III. All points with a negative y-coordinate are in the solution set of the inequality.

(A) I only
(B) II only
(C) I and II
(D) II and III

41 Which of the following is the factored form of the polynomial $3x^2 - 150m + 25mx - 18x$?

(A) $(x - 6)(3x + 25m)$
(B) $(x + 6)(3x - 25m)$
(C) $(x + m)(6 - 25m)$
(D) $(x + m)(6 + 25m)$

42 Given $A(x) = 9x^2 - ax + 1$ and $B(x) = x^2 + bx + 25$, where a and b are positive real numbers, and if $A(x)$ and $B(x)$ represent perfect squares by having equal roots, what is the value of $a + b$?

(A) 2
(B) 8
(C) 16
(D) 24

43 The sum of two numbers is 41 and their product is 408. The largest number is:

(A) 8
(B) 12
(C) 17
(D) 24

44 **Question 44.** What is the value of the radicand in the simplified form of $\sqrt[3]{8a^4b^3} \times \sqrt[3]{ab^5}$ if $a = 2$ and $b = -1$?

A. -4
B. 4
C. 8
D. 16

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46 The perimeter of a triangle is 126 cm. The sides of this triangle are in the ratio of $4 : 6 : 8$. What is the measure of the longest side of this triangle?

(A) 28 cm
(B) 35 cm
(C) 42 cm
(D) 56 cm

47 Zein bought a sofa at a 15% discount. After a year, he sold it for half the price he had paid, receiving $\$637.50$. What was the original price of the sofa?

(A) $1,400
(B) $1,500
(C) $1,700
(D) $1,800

48 A shirt originally costs $\$40$. It was discounted by 10% the first time, then by 20%. What is the final price of the shirt?

(A) $28.80
(B) $29.40
(C) $34.50
(D) $35.60

49 Mazen consumed 600 calories at breakfast and $1,200$ calories at lunch. He then went to the gym and burned 350 calories. If his daily net calorie intake should be $1,600$ calories, how many calories should he consume at dinner?

(A) 100
(B) 150
(C) 200
(D) 350

50 **Question 50.** Which of the following statements is true about $A(x) = \frac{2x}{(3x-1)^2}$?

A. It is always negative for all real numbers x.
B. It is always positive for all real numbers x.
C. It is equal to zero when $x = 0$ or $x = \frac{1}{3}$.
D. It is negative for all values of $x < 0$.

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Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	A	26	C
2	C	27	D
3	B	28	A
4	B	29	B
5	A	30	D
6	C	31	D
7	A	32	C
8	D	33	B
9	A	34	A
10	B	35	A
11	C	36	B
12	B	37	B
13	B	38	B
14	B	39	C
15	D	40	B
16	B	41	A
17	A	42	C
18	A	43	D
19	D	44	B
20	C	45	C
21	A	46	D
22	B	47	B
23	C	48	A
24	D	49	B
25	D	50	D

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Detailed Solutions

50 SOLUTIONS

1 **Answer: A**

See the answer key.

2 **Answer: C** 98

Total number of tests across all three terms is $3 + 2 + 1 = 6$.
To have an overall average of at least 90, the total sum of the 6 scores must be at least:

$$6 \times 90 = 540$$

Current sum of scores from Term 1 and Term 2:

$$\text{Term } 1 = 3 \times 88 = 264$$

$$\text{Term } 2 = 90 + 88 = 178$$

$$\text{Current Total} = 264 + 178 = 442$$

Let x be the score on the final test:

$$442 + x \geq 540 \implies x \geq 540 - 442 = 98$$

Therefore, the minimum score required is 98.

3 **Answer: B**

See the answer key.

4 **Answer: B**

See the answer key.

5 **Answer: A**

See the answer key.

6 **Answer: C**

See the answer key.

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7 **Answer: A**

See the answer key.

8 **Answer: D**

See the answer key.

9 **Answer: A**

See the answer key.

10 **Answer: B**

See the answer key.

11 **Answer: C**

See the answer key.

12 **Answer: B**

See the answer key.

13 **Answer: B** The bar graph shows a general decrease in sales from 2022 to 2023.

Total sales in 2022 = 400+600+300+200 = 1500. Total sales in 2023 = 100+400+500+200 = 1200. There is a general decrease.

14 **Answer: B**

See the answer key.

15 **Answer: D**

See the answer key.

16 **Answer: B**

See the answer key.

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17 **Answer: A** $\frac{x+y}{2}$

Let $x = 6$ and $y = 4$ (both are even natural numbers):
- $\frac{x+y}{2} = \frac{6+4}{2} = 5$, which is an odd integer, and $5 < 6 = x$.
- $2x - y$ is always even (even minus even is even).
- $x + \frac{y}{2} > x$ for all natural numbers y.
- $x + y$ is always even and greater than x.

Therefore, only $\frac{x+y}{2}$ can be an odd integer less than x.

18 **Answer: A** $\frac{30y-x}{17}$

Multiply the entire equation by 2:

$$6r + x + r = 10(3y - r)$$

$$7r + x = 30y - 10r$$

Add $10r$ to both sides and subtract x:

$$17r = 30y - x \implies r = \frac{30y - x}{17}$$

19 **Answer: D** $(-\frac{2}{9}, \frac{17}{18})$

The line f passes through $(-1, -1)$ and $(0, 1.5)$:

$$\text{Slope } m = \frac{1.5 - (-1)}{0 - (-1)} = 2.5 = \frac{5}{2}$$

So the equation of line f is:

$$f(x) = \frac{5}{2}x + \frac{3}{2}$$

Equating $f(x) = g(x)$:

$$\frac{5}{2}x + \frac{3}{2} = -2x + \frac{1}{2}$$

Multiply by 2:

$$5x + 3 = -4x + 1 \implies 9x = -2 \implies x = -\frac{2}{9}$$

Now find y:

$$y = -2\left(-\frac{2}{9}\right) + \frac{1}{2} = \frac{4}{9} + \frac{1}{2} = \frac{8+9}{18} = \frac{17}{18}$$

Thus, the point of intersection is $(-\frac{2}{9}, \frac{17}{18})$.

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20 **Answer: C** I and III

Expand $(i-3)(3i-1)$:

$$(i-3)(3i-1) = 3i^2 - i - 9i + 3 = 3(-1) - 10i + 3 = -3 - 10i + 3 = -10i$$

Therefore:

$$A = 2i(-10i) \quad \text{(Statement I is equivalent)}$$

Now compute A:

$$A = 2i(-10i) = -20i^2 = -20(-1) = 20 \quad \text{(Statement III is equivalent)}$$

Statement II is incorrect. Thus, Statements I and III are equivalent to A.

21 **Answer: A** It has a vertical asymptote at $x = 3$.

For $f(x) = \sqrt{3x-9}$:
- The radicand requires $3x - 9 \geq 0 \implies x \geq 3$, so the domain is $[3, +\infty)$.
- At $x = 3$, $f(3) = \sqrt{0} = 0$, so $(3, 0)$ is on the graph.
- The range is $[0, +\infty)$.
- A square root function is continuous on its domain and does NOT have a vertical asymptote at $x = 3$.

Therefore, 'It has a vertical asymptote at $x = 3$' is the statement that is NOT correct.

22 **Answer: B** $(0.5, 0)$

The absolute value expression $|2x - 1|$ is always ≥ 0, and achieves its minimum value of 0 when $2x - 1 = 0 \implies x = 0.5$.
At $x = 0.5$, $g(0.5) = 2|0| = 0$.
Thus, the local minimum is at $(0.5, 0)$.

23 **Answer: C** $2,464\ \text{cm}^2$

The side of the square cardboard is $\frac{200}{4} = 50$ cm.
After cutting corners of size x, the base of the open box has dimensions $(50 - 2x) \times (50 - 2x)$ and height x.

$$V = x(50 - 2x)^2 = 5,808$$

Testing $x = 3$:

$$3(50 - 6)^2 = 3(44^2) = 3(1,936) = 5,808\ \text{cm}^3$$

So $x = 3$ is a solution.
The surface area of the resulting open-top box is the original cardboard area minus the area of the 4 cut-out corner squares:

$$\text{Surface Area} = 50^2 - 4(x^2) = 2,500 - 4(3^2) = 2,500 - 36 = 2,464\ \text{cm}^2$$

24 **Answer: D**

See the answer key.

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25 **Answer: D** Graph D

Simplify the rational function:

$$f(x) = \frac{2}{2(x+2)} + 1 = \frac{1}{x+2} + 1$$

Key characteristics:
- **Vertical asymptote:** $x + 2 = 0 \implies x = -2$.
- **Horizontal asymptote:** $y = 1$.
- **y-intercept:** $f(0) = \frac{1}{2} + 1 = 1.5$.
- **x-intercept:** $0 = \frac{1}{x+2} + 1 \implies \frac{1}{x+2} = -1 \implies x = -3$.
- **Branches:** Right branch is for $x > -2$ (above $y = 1$); left branch is for $x < -2$ (below $y = 1$).

Examining the options, Graph A matches these exact features.

26 **Answer: C**

See the answer key.

27 **Answer: D** 19

The two angles are $(x + 3)^\circ$ and $2(x + 3)^\circ$.
Their sum is:

$$(x + 3) + 2(x + 3) = 66 \implies 3(x + 3) = 66$$

Divide by 3:

$$x + 3 = 22 \implies x = 19$$

28 **Answer: A** $x_1 = \frac{-4-\sqrt{14}}{2}$

By Vieta's formulas, the sum of the roots of $2x^2 + ax + 1 = 0$ is:

$$x_1 + x_2 = -\frac{a}{2} = -4 \implies a = 8$$

Now solve $2x^2 + 8x + 1 = 0$ using the quadratic formula:

$$x = \frac{-8 \pm \sqrt{8^2 - 4(2)(1)}}{2(2)} = \frac{-8 \pm \sqrt{64 - 8}}{4} = \frac{-8 \pm \sqrt{56}}{4} = \frac{-8 \pm 2\sqrt{14}}{4} = \frac{-4 \pm \sqrt{14}}{2}$$

Since $x_1 < x_2$, we take the minus sign:

$$x_1 = \frac{-4 - \sqrt{14}}{2}$$

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29 **Answer: B** 18.75

Inverse variation with x^2 means:

$$y = \frac{k}{x^2} \implies k = yx^2$$

Using $y = 12$ and $x = 5$:

$$k = 12 \times 5^2 = 12 \times 25 = 300$$

When $x = 4$:

$$y = \frac{300}{4^2} = \frac{300}{16} = 18.75$$

30 **Answer: D** $4\pi \text{ cm}^2$

The area of a sector with radius r and arc length L is given by:

$$\text{Area} = \frac{1}{2}rL$$

Given $r = 4 \text{ cm}$ and $L = 2\pi \text{ cm}$:

$$\text{Area} = \frac{1}{2}(4)(2\pi) = 4\pi \text{ cm}^2$$

31 **Answer: D**

See the answer key.

32 **Answer: C**

See the answer key.

33 **Answer: B**

See the answer key.

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34 **Answer: A** 9

First, calculate the slope of line AM:

$$m = \frac{10-4}{3-(-3)} = \frac{6}{6} = 1$$

The equation of line AM using point $M(3,10)$ is:

$$y - 10 = 1(x-3) \implies y = x + 7$$

Since $H(2,b)$ lies on this line:

$$b = 2 + 7 = 9$$

35 **Answer: A**

See the answer key.

36 **Answer: B**

See the answer key.

37 **Answer: B**

See the answer key.

38 **Answer: B** $650

Total money spent over 5 days:

$$\$3,400 - \$150 = \$3,250$$

Average spent per day:

$$\frac{\$3,250}{5} = \$650$$

39 **Answer: C** 750

Convert kilometers per hour to meters per minute:

$$1\text{ km} = 1,000\text{ m}, \quad 1\text{ h} = 60\text{ min}$$

$$45\text{ km/h} = \frac{45 \times 1,000}{60}\text{ m/min} = \frac{45,000}{60} = 750\text{ m/min}$$

Thus, $a = 750$.

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40 **Answer: B** II only

Rewrite the inequality as $y \leq -3x - 2$:
- **Statement I:** $y \leq -3x - 2$ represents the region *on or below* the line, not above (Statement I is false).
- **Statement II:** Substitute $(1, -5)$: $-5 + 3(1) = -2 \leq -2$, which is true. The point lies on the boundary line (Statement II is true).
- **Statement III:** Consider the point $(5, -1)$ which has a negative y-coordinate: $-1 + 3(5) = 14 \not\leq -2$, so not all points with negative y-coordinates are solutions (Statement III is false).

Therefore, only Statement II is true.

41 **Answer: A** $(x - 6)(3x + 25m)$

Group terms with common factors:

$$3x^2 - 18x + 25mx - 150m = 3x(x - 6) + 25m(x - 6)$$

Factor out $(x - 6)$:

$$(x - 6)(3x + 25m)$$

42 **Answer: C** 16

A quadratic $Ax^2 + Bx + C$ has equal roots when its discriminant is zero:
- For $A(x) = 9x^2 - ax + 1$:

$$\Delta = (-a)^2 - 4(9)(1) = 0 \implies a^2 = 36 \implies a = 6 \quad (a > 0)$$

- For $B(x) = x^2 + bx + 25$:

$$\Delta = b^2 - 4(1)(25) = 0 \implies b^2 = 100 \implies b = 10 \quad (b > 0)$$

Therefore:

$$a + b = 6 + 10 = 16$$

43 **Answer: D** 24

Let the two numbers be the roots of $t^2 - 41t + 408 = 0$.
Using the quadratic formula:

$$t = \frac{41 \pm \sqrt{41^2 - 4(408)}}{2} = \frac{41 \pm \sqrt{1,681 - 1,632}}{2} = \frac{41 \pm \sqrt{49}}{2} = \frac{41 \pm 7}{2}$$

The two numbers are $\frac{48}{2} = 24$ and $\frac{34}{2} = 17$.
The largest number is 24.

44 **Answer: B**

See the answer key.

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45 **Answer: C**

See the answer key.

46 **Answer: D** 56 cm

Let the side lengths be $4k$, $6k$, and $8k$.
Perimeter:

$$4k + 6k + 8k = 126 \implies 18k = 126 \implies k = 7$$

The longest side is $8k$:

$$8 \times 7 = 56 \text{ cm}$$

47 **Answer: B** $1,500

Let the original price be P.
Zein bought it at a 15% discount, so he paid $0.85P$.
He sold it for half of that amount, receiving $\$637.50$:

$$\frac{0.85P}{2} = 637.50 \implies 0.85P = 1,275 \implies P = \frac{1,275}{0.85} = \$1,500$$

48 **Answer: A** $28.80

After the first discount of 10%, the price is:

$$40 \times (1 - 0.10) = 40 \times 0.90 = \$36$$

After the second discount of 20%, the final price is:

$$36 \times (1 - 0.20) = 36 \times 0.80 = \$28.80$$

49 **Answer: B** 150

Let d be the calories Mazen consumes at dinner.
His net calorie intake is:

$$\text{Net Calories} = 600 + 1,200 + d - 350 = 1,450 + d$$

Setting this equal to the goal of $1,600$ calories:

$$1,450 + d = 1,600 \implies d = 150 \text{ calories}$$

50 **Answer: D**

See the answer key.

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